

1. Overview

This folder contains the audit trail for the identification phase of the SLR. To support reproducibility, it documents the exact Boolean search strings, database filters, search date, and raw metadata for the **639** records initially identified.

2. Detailed Search Strings

The search strategy was executed on March 16, 2026, across five academic databases: Web of Science, Scopus, IEEE Xplore, ScienceDirect, and ACM Digital Library. The search strings were designed to capture studies at the intersection of insider threat detection and data-centric machine learning, including preprocessing, feature engineering, feature extraction, and behavioral indicators..

SEARCH STRINGS AND NUMBER OF RECORDS RETRIEVED.		
Database	Search String	Results
Web of Science	("Artificial Intelligence" OR "Machine Learning" OR "Deep Learning" OR "Neural Networks") AND ("Feature Engineering" OR "Feature Extraction" OR "Feature Selection" OR "Behavior Indicators" OR "Behavioral Features") AND ("Insider Threat Detection" OR "Insider Threat Detection System" OR "Insider Threat" OR "Cyber Threats" OR "Computer Threats" OR "Cybersecurity" OR "Security Threats") AND ("User Behavior" OR "User Behaviour" OR "Behavioral Detection" OR "Behavioral-based Detection" OR "User Behavior-based Detection" OR "Behavioral")	181
Scopus	TITLE-ABS-KEY (("insider threat*" OR "malicious insider") AND ("feature engineer*" OR "feature extract*" OR "feature selection" OR "representation learning" OR "behavior* indicator*" OR "preprocess*" OR "machine learning" OR "deep learning" OR "anomaly detection")) AND PUBYEAR > 2020 AND PUBYEAR < 2026 AND (LIMIT-TO (DOCTYPE , "ar")) AND (LIMIT-TO (LANGUAGE , "English"))	174
IEEE Xplore	("insider threat") AND ("machine learning" OR "deep learning" OR "anomaly detection") AND ("feature engineering" OR "feature extraction" OR "representation learning" OR "preprocessing" OR "behavioral indicators")	116
Science Direct	("insider threat") AND ("feature engineering" OR "representation learning") AND ("machine learning" OR "deep learning") AND ("log" OR "logs")	120
ACM Digital Library	"query": ("insider threat" AND ("machine learning" OR "deep learning" OR "anomaly detection") AND ("feature engineering" OR "feature extraction" OR "representation learning" OR "preprocessing" OR "behavioral indicators")) "filter": Article Type: Research Article, Survey, E-Publication Date: (01/01/2021 TO 12/31/2025)	48
Total Records Identified		639

2. Files with Raw Results

To allow for independent verification, the table below maps the results to the specific files stored in this directory. All files were exported on March 16, 2026.

Database	Result Count	Primary Export Filename	Format
Web of Science	181	WoS_181_Records_Retrieved_16_March_2026	CSV
Scopus	174	scopus_174_records_export_Mar 16	CSV
ScienceDirect	120	Science Direct 120 Records	CSV
IEEE Xplore	116	IEEE 116 Records	CSV
ACM Digital Library	48	acm 48 Records 16 Mar 2026	CSV
Total	639	-	-

3. Multi-Stage Filtration Summary

Following the initial identification of **639** records, we applied a systematic filtration process to arrive at the final **110** primary studies. This process was conducted in accordance with **PRISMA 2020** standards and guidelines.

Database	Stage 1 (Full Search Count)		Stage 2 Removing duplicates	Stage 3 (Title and Abstract)	Stage 4 (Full Text)	Final Count
	Initial Results	Filtered by Year (2021- 2025 Journals Only				
Web of Science	181	118	350	124	117	110
Scopus	174	174				
Science Direct	120	89				
IEEE Xplore	116	11				
ACM Digital Library	48	5				
Remaining Papers	639	397	350	124	110	110
Excluded	242		47	226	7	

As shown in Table 3, records outside the 2021–2025 period, non-English papers, and non-journal publications were removed during database filtering.

Duplicate records were then removed before title and abstract screening. During screening, papers were excluded if they were not directly related to ML-based insider threat detection, did not report relevant datasets, or lacked sufficient discussion of preprocessing or feature engineering.

Full-text eligibility and quality assessment were then applied to identify the final 110 studies that provided enough technical evidence to answer RQ1–RQ4.

The next folder, 02_Study_Selection, provides further details on the screening and selection process following the PRISMA methodology..